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CS 362 Lab 6

3/3/2026

Windows Forensics Part 1

Part 1: Non-technical summary

In this lab, we focused on Windows operating system forensics. The first step in Windows forensics methodology is to collect volatile information. Following this structure, we began by gathering basic data from a running Windows system before anything could change or disappear. The first few pieces of volatile information we collected with the system's current data and time to create a timeline. The next piece of information we gathered was the list of logged-on users. This step was used to check who or what was actively using the system. In order to display this list in several ways, we added four different options to the original command to better understand the information.

We then examined the network connections using multiple command options to detect any active communication with other systems. After that, we focused on collecting process information to understand which programs and processes were active in memory. After inputting the various commands, we reviewed the process details. This allowed us to gain a clearer picture of the system's activity. The last task we touched on was exploring the Windows print spool to see how print jobs are temporarily stored. Overall, this lab helped us to gain a better understanding of how to collect and analyze volatile information during a forensic investigation.

Part 2: Technical summary

To begin this lab, we started with the basics of collecting volatile information: collecting the time and date of the device being used. I used the command **date /t & time /t** to perform this task.

```
C:\Windows\system32>date /t & time /t  
Tue 02/24/2026  
02:37 PM
```

This step is important when needing to extract a timeline for an incident. The next piece of volatile information we looked at is the list of logged-on users. In order to use the command to access the logged-on users, we first need to download Sysinternals Suite. We do so by entering <https://download.sysinternals.com/files/SysinternalsSuite.zip> into a browser, and the folder downloads immediately. After the folder was successfully downloaded, I extracted all the files from the zip file into the Documents folder. I then changed the working directory using the command **cd C:\Users\vboxuser\Documents\SysinternalsSuite** to be able to access the Sysinternal Suite folder. I input the following command to finally list all logged-on users: **psloggedon**.

```
C:\Users\vboxuser\Documents\SysinternalsSuite>psloggedon

PsLoggedon v1.35 - See who's logged on
Copyright (C) 2000-2016 Mark Russinovich
Sysinternals - www.sysinternals.com

Users logged on locally:
      2/24/2026 3:07:21 PM      WINDOWS\vboxuser

No one is logged on via resource shares.
```

In the next step of this lab, we used the command **logonsessions** | **findstr "logon session"** to list the logged-on sessions of this device. Adding **findstr** to this command lists only the session IDs.

```
C:\Users\vboxuser\Documents\SysinternalsSuite>logonsessions | findstr "logon session"
LogonSessions v1.41 - Lists logon session information
[0] Logon session 00000000:000003e7:
[1] Logon session 00000000:00006a4c:
[2] Logon session 00000000:00006e43:
[3] Logon session 00000000:00006e63:
[4] Logon session 00000000:00003e4:
[5] Logon session 00000000:0000be52:
[6] Logon session 00000000:0000be99:
[7] Logon session 00000000:00003e5:
[8] Logon session 00000000:0001b5a9:
[9] Logon session 00000000:0001b5e8:
```

If we use the **-p** option along with the command **logonsessions**, we can display a list of processes running in each session. Because the list was very lengthy, I only added screenshots of the beginning and end of the output.

```
C:\Users\vboxuser\Documents\SysinternalsSuite>logonsessions -p
LogonSessions v1.41 - Lists logon session information
Copyright (C) 2004-2020 Mark Russinovich
Sysinternals - www.sysinternals.com

[0] Logon session 00000000:000003e7:
    User name:  WORKGROUP\WINDOWS$
    Auth package:  NTLM
    Logon type:  (none)
    Session:  0
    Sid:  S-1-5-18
    Logon time:  2/24/2026 3:06:44 PM
    Logon server:
    DNS Domain:
    UPN:
        608: winlogon.exe
        664: lsass.exe
        764: svchost.exe
        368: svchost.exe
        424: svchost.exe
        1936: svchost.exe
        2012: spoolsv.exe
        2264: svchost.exe
        2740: MicrosoftEdgeUpdate.exe
        2292: dllhost.exe
        3392: SearchIndexer.exe
        2228: MoUsoCoreWorker.exe
```

```
5720: conhost.exe
4332: MpCmdRun.exe
3380: conhost.exe
2800: mpam-d.exe

[1] Logon session 00000000:00006a4c:
    User name:
    Auth package:  NTLM
    Logon type:  (none)
    Session:  0
    Sid:  (none)
    Logon time:  2/24/2026 3:06:45 PM
    Logon server:
    DNS Domain:
    UPN:

[2] Logon session 00000000:00006e43:
    User name:  Font Driver Host\UMFD-0
    Auth package:  Negotiate
    Logon type:  Interactive
    Session:  0
    Sid:  S-1-5-96-0-0
    Logon time:  2/24/2026 3:06:46 PM
    Logon server:
    DNS Domain:
    UPN:
        792: fontdrvhost.exe

[3] Logon session 00000000:00006e63:
    User name:  Font Driver Host\UMFD-1
```

```
[9] Logon session 00000000:0001b5e8:
User name: WINDOWS\vboxuser
Auth package: NTLM
Logon type: Interactive
Session: 1
Sid: S-1-5-21-2832613360-3222320947-3522146261-1000
Logon time: 2/24/2026 3:07:12 PM
Logon server: WINDOWS
DNS Domain:
UPN:
2648: sihost.exe
2696: svchost.exe
2748: taskhostw.exe
2784: taskhostw.exe
3016: ctfmon.exe
3528: explorer.exe
2860: svchost.exe
2920: TextInputHost.exe
4088: StartMenuExperienceHost.exe
2232: RuntimeBroker.exe
4356: RuntimeBroker.exe
5048: dllhost.exe
3520: SecurityHealthSystray.exe
3608: M365Copilot.exe
5000: RuntimeBroker.exe
4236: msedgewebview2.exe
892: msedgewebview2.exe
1640: msedgewebview2.exe
3880: msedgewebview2.exe
3108: msedgewebview2.exe
```

```
4128: UserOOBEBroker.exe
548: SearchApp.exe
4420: msedgewebview2.exe
5412: backgroundTaskHost.exe
6940: RuntimeBroker.exe
5804: ShellExperienceHost.exe
7144: msedge.exe
3860: msedge.exe
5288: msedge.exe
340: msedge.exe
7120: msedge.exe
3680: RuntimeBroker.exe
376: HxTsr.exe
3244: RuntimeBroker.exe
3840: OneDrive.exe
6352: OneDrive.Sync.Service.exe
```

There are two other variations of the **logonsessions** command we can use to display different presentations of the logged-on sessions. When using the following command, the list of sessions is shown as comma-separated values: **logonsessions -c**.

```
C:\Users\vboxuser\Documents\SysinternalsSuite>logonsessions -c

LogonSessions v1.41 - Lists logon session information
Copyright (C) 2004-2020 Mark Russinovich
Sysinternals - www.sysinternals.com

Logon Session,User Name,Auth Package,Logon Type,Session,Sid,Logon Time,Logon Server,DNS Domain,UPN,Processes
00000000:000003e7,WORKGROUP\WINDOWS$,NTLM,(none),0,S-1-5-18,2/24/2026 3:06:44 PM,,,,,
00000000:00006a4c,,NTLM,(none),0,(none),2/24/2026 3:06:45 PM,,,,,
00000000:00006e43,Font Driver Host\UMFD-0,Negotiate,Interactive,0,S-1-5-96-0-0,2/24/2026 3:06:46 PM,,,,,
00000000:00006e63,Font Driver Host\UMFD-1,Negotiate,Interactive,1,S-1-5-96-0-1,2/24/2026 3:06:46 PM,,,,,
00000000:00003e4,WORKGROUP\WINDOWS$,Negotiate,Service,0,S-1-5-20,2/24/2026 3:06:46 PM,,,,,
00000000:0000be52,Window Manager\DWM-1,Negotiate,Interactive,1,S-1-5-90-0-1,2/24/2026 3:06:48 PM,,,,,
00000000:0000be99,Window Manager\DWM-1,Negotiate,Interactive,1,S-1-5-90-0-1,2/24/2026 3:06:48 PM,,,,,
00000000:000003e5,NT AUTHORITY\LOCAL SERVICE,Negotiate,Service,0,S-1-5-19,2/24/2026 3:06:49 PM,,,,,
00000000:0001b5a9,WINDOWS\vboxuser,NTLM,Interactive,1,S-1-5-21-2832613360-3222320947-3522146261-1000,2/24/2026 3:07:12 PM,WINDOWS,,,
00000000:0001b5e8,WINDOWS\vboxuser,NTLM,Interactive,1,S-1-5-21-2832613360-3222320947-3522146261-1000,2/24/2026 3:07:12 PM,WINDOWS,,,

```

The second variation of this command is **logonsessions -ct**. This command shows the list of sessions as tab-delimited values. The columns list the logon session, username, authentication package, logon type, session ID, and login time for each session.

```
C:\Users\vboxuser\Documents\SysinternalsSuite>logonsessions -ct

LogonSessions v1.41 - Lists logon session information
Copyright (C) 2004-2020 Mark Russinovich
Sysinternals - www.sysinternals.com
```

Logon Session	User Name	Auth Package	Logon Type	Session Sid	Logon Time
00000000:00003e7	WORKGROUP\WINDOWS\$	NTLM	(none)	0	S-1-5-18
00000000:00006a4c		NTLM	(none)	0	2/24/2026 3:06:45 PM
00000000:00006e43	Font Driver Host\UMFD-0	Negotiate	Interactive	0	S-1-5-96-0-0
00000000:00006e63	Font Driver Host\UMFD-1	Negotiate	Interactive	1	S-1-5-96-0-1
00000000:00003e4	WORKGROUP\WINDOWS\$	Negotiate	Service	0	S-1-5-20
00000000:0000be52	Window Manager\DWM-1	Negotiate	Interactive	1	S-1-5-90-0-1
00000000:0000be99	Window Manager\DWM-1	Negotiate	Interactive	1	S-1-5-90-0-1
00000000:00003e5	NT AUTHORITY\LOCAL SERVICE	Negotiate	Service	0	S-1-5-19
00000000:0001b5a9	WINDOWS\vboxuser	NTLM	Interactive	1	S-1-5-21-283
2613360-3222320947-3522146261-1000	WINDOWS\vboxuser	NTLM	Interactive	1	S-1-5-21-283

Another form of volatile information is the network connection of the device. There are many variations of this command, but the most basic version is to use **netstat -a** in order to list all the connections.

```
C:\Users\vboxuser\Documents\SysinternalsSuite>netstat -a

Active Connections
```

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	Windows:0	LISTENING
TCP	0.0.0.0:445	Windows:0	LISTENING
TCP	0.0.0.0:5040	Windows:0	LISTENING
TCP	0.0.0.0:7680	Windows:0	LISTENING
TCP	0.0.0.0:49664	Windows:0	LISTENING
TCP	0.0.0.0:49665	Windows:0	LISTENING
TCP	0.0.0.0:49666	Windows:0	LISTENING
TCP	0.0.0.0:49667	Windows:0	LISTENING
TCP	0.0.0.0:49669	Windows:0	LISTENING
TCP	0.0.0.0:49674	Windows:0	LISTENING
TCP	10.0.2.15:139	Windows:0	LISTENING
TCP	10.0.2.15:50092	172.183.7.192:https	ESTABLISHED
TCP	10.0.2.15:50120	a23-60-78-147:http	TIME_WAIT
TCP	10.0.2.15:50121	a23-213-53-15:https	ESTABLISHED
TCP	10.0.2.15:50125	a23-32-45-19:http	ESTABLISHED
TCP	10.0.2.15:50128	150.171.27.11:https	ESTABLISHED
TCP	10.0.2.15:50129	150.171.27.11:https	ESTABLISHED
TCP	[::]:135	Windows:0	LISTENING
TCP	[::]:445	Windows:0	LISTENING
TCP	[::]:7680	Windows:0	LISTENING
TCP	[::]:49664	Windows:0	LISTENING
TCP	[::]:49665	Windows:0	LISTENING
TCP	[::]:49666	Windows:0	LISTENING
TCP	[::]:49667	Windows:0	LISTENING


```

TCP    [::]:49669          Windows:0           LISTENING
TCP    [::]:49674          Windows:0           LISTENING
TCP    [::1]:42050         Windows:0           LISTENING
UDP    0.0.0.0:5050        *:                  *
UDP    0.0.0.0:5353        *:                  *
UDP    0.0.0.0:5355        *:                  *
UDP    10.0.2.15:137       *:                  *
UDP    10.0.2.15:138       *:                  *
UDP    10.0.2.15:1900      *:                  *
UDP    10.0.2.15:60136     *:                  *
UDP    127.0.0.1:1900      *:                  *
UDP    127.0.0.1:60137     *:                  *
UDP    127.0.0.1:63087     *:                  *
UDP    [::]:5353           *:                  *
UDP    [::]:5355           *:                  *
UDP    [::1]:1900          *:                  *
UDP    [::1]:60135         *:                  *
UDP    [fe80::57ea:387f:5feb:86ec%5]:1900 *:                  *
UDP    [fe80::57ea:387f:5feb:86ec%5]:60134 *:                  *

```

The second variation of the netstat command is **netstat -p protocol**. This command lists the connections by the protocol you input from these options: TCP, UDP, TCPv6, and UDPv6.

```

C:\Users\vbouser\Documents\SysinternalsSuite>netstat -p TCP
Active Connections
  Proto Local Address           Foreign Address         State
  TCP   10.0.2.15:50092         172.183.7.192:https    ESTABLISHED
  TCP   10.0.2.15:50130         a23-213-53-15:https    CLOSE_WAIT
  TCP   10.0.2.15:50136         13.107.246.51:https    CLOSE_WAIT
  TCP   10.0.2.15:50149         51.11.192.49:https     TIME_WAIT

C:\Users\vbouser\Documents\SysinternalsSuite>netstat -p UDP
Active Connections
  Proto Local Address           Foreign Address         State

C:\Users\vbouser\Documents\SysinternalsSuite>netstat -p TCPv6
Active Connections
  Proto Local Address           Foreign Address         State

C:\Users\vbouser\Documents\SysinternalsSuite>netstat -p UDPv6
Active Connections
  Proto Local Address           Foreign Address         State

```

The third option to use with the **netstat** command is **-r**. This variation displays the routing table of the active routes happening in the device being used.

```

C:\Users\vboxuser\Documents\SysinternalsSuite>netstat -r
=====
Interface List
  5...08 00 27 b6 88 38 .....Intel(R) PRO/1000 MT Desktop Adapter
  1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          10.0.2.2         10.0.2.15        25
10.0.2.0                  255.255.255.0    On-link          10.0.2.15        281
10.0.2.15                  255.255.255.255  On-link          10.0.2.15        281
10.0.2.255                 255.255.255.255  On-link          10.0.2.15        281
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255            255.255.255.255  On-link          127.0.0.1        331
224.0.0.0                  240.0.0.0        On-link          127.0.0.1        331
224.0.0.0                  240.0.0.0        On-link          10.0.2.15        281
255.255.255.255            255.255.255.255  On-link          127.0.0.1        331
255.255.255.255            255.255.255.255  On-link          10.0.2.15        281
=====

Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination      Gateway
5 281 ::/0 fe80::2
1 331 ::1/128 On-link
5 281 fd17:625c:f037:2::/64 On-link
5 281 fd17:625c:f037:2:1d5f:e376:657b:c0da/128 On-link
5 281 fd17:625c:f037:2:5938:3270:f813:520e/128 On-link
5 281 fe80::/64 On-link
5 281 fe80::57ea:387f:5feb:86ec/128 On-link

```

The fourth variation is

netstat -b. This command displays the executable involved in creating each connection.

```

C:\Users\vboxuser\Documents\SysinternalsSuite>netstat -b

Active Connections

Proto Local Address           Foreign Address         State
TCP    10.0.2.15:50092         172.183.7.192:https     ESTABLISHED
ShellHWDetection
[svchost.exe]
TCP    10.0.2.15:50130         a23-213-53-15:https     CLOSE_WAIT
[SearchApp.exe]
TCP    10.0.2.15:50136         13.107.246.51:https     CLOSE_WAIT
[SearchApp.exe]

```

The last option we visit

for the **netstat** command is **-e**. This command allows us to display the interface statistics that were received and sent.

```
C:\Users\vboxuser\Documents\SysinternalsSuite>netstat -e
Interface Statistics
```

	Received	Sent
Bytes	3504588	325288
Unicast packets	3736	1644
Non-unicast packets	12	12
Discards	0	0
Errors	0	0
Unknown protocols	0	0

To collect process information, we can use the command **tasklist**, **listdlls**, **handle**, and **pslist**. The **tasklist** command lists the image name, PID, session name, session number, and memory usage of each task. Because the list was very lengthy, I only included the beginning of the output in the screenshot.

```
C:\Users\vboxuser\Documents\SysinternalsSuite>tasklist
```

Image Name	PID	Session Name	Session#	Mem Usage
System Idle Process	0	Services	0	8 K
System	4	Services	0	152 K
Registry	92	Services	0	47,648 K
smss.exe	324	Services	0	960 K
csrss.exe	420	Services	0	4,660 K
wininit.exe	512	Services	0	6,136 K
csrss.exe	520	Console	1	5,936 K
winlogon.exe	608	Console	1	9,428 K
services.exe	644	Services	0	7,256 K
lsass.exe	664	Services	0	16,320 K
svchost.exe	764	Services	0	30,476 K
fontdrvhost.exe	784	Console	1	4,244 K
fontdrvhost.exe	792	Services	0	2,340 K
svchost.exe	876	Services	0	14,128 K
dwm.exe	988	Console	1	62,208 K
svchost.exe	368	Services	0	72,704 K
svchost.exe	424	Services	0	88,356 K
svchost.exe	720	Services	0	23,528 K
svchost.exe	704	Services	0	25,708 K
svchost.exe	1160	Services	0	28,984 K
Memory Compression	1404	Services	0	25,900 K
svchost.exe	1436	Services	0	19,000 K
svchost.exe	1508	Services	0	10,220 K
svchost.exe	1644	Services	0	12,456 K
svchost.exe	1768	Services	0	10,216 K
svchost.exe	1784	Services	0	8,128 K
svchost.exe	1936	Services	0	19,176 K
spoolsv.exe	2012	Services	0	10,848 K
svchost.exe	1288	Services	0	14,100 K
svchost.exe	1756	Services	0	7,348 K
svchost.exe	2264	Services	0	37,736 K
MpDefenderCoreService.exe	2336	Services	0	18,732 K
MsMpEng.exe	2368	Services	0	156,288 K

When trying to conduct a narrower search using the **tasklist** command, we can filter the processes based on a specific parameter of either the PID, image name, session name, session number, or memory usage. For example, we used the command **tasklist /FI "PID gt 700"** to filter the processes by PID. The screenshot includes only the beginning of the output because the list was extensive.

```
C:\Users\vboxuser\Documents\SysinternalsSuite>tasklist /FI "PID gt 700"
```

Image Name	PID	Session Name	Session#	Mem Usage
svchost.exe	764	Services	0	29,780 K
fontdrvhost.exe	784	Console	1	3,948 K
fontdrvhost.exe	792	Services	0	2,360 K
svchost.exe	876	Services	0	14,104 K
dwm.exe	988	Console	1	62,052 K
svchost.exe	720	Services	0	22,972 K
svchost.exe	704	Services	0	21,480 K
svchost.exe	1160	Services	0	28,272 K
Memory Compression	1404	Services	0	59,364 K
svchost.exe	1436	Services	0	22,908 K
svchost.exe	1508	Services	0	10,252 K
svchost.exe	1644	Services	0	12,508 K
svchost.exe	1768	Services	0	9,916 K
svchost.exe	1784	Services	0	8,016 K
svchost.exe	1936	Services	0	18,032 K
spoolsv.exe	2012	Services	0	10,892 K
svchost.exe	1288	Services	0	12,576 K
svchost.exe	1756	Services	0	7,368 K
svchost.exe	2264	Services	0	37,096 K
MpDefenderCoreService.exe	2336	Services	0	16,428 K
MsMpEng.exe	2368	Services	0	164,628 K
sihost.exe	2648	Console	1	25,096 K

The **listdlls** command

displayed another long and detailed output. I included the very beginning of it in the screenshot. This list included the base name, size, and path of the processes.

```
Users\vboxuser\Documents\SysinternalsSuite>listdlls

tdlls v3.2 - Listdlls
yright (C) 1997-2016 Mark Russinovich
internals

or opening System(4):
ess is denied.

or opening Registry(92):
ess is denied.

or opening smss.exe(324):
ess is denied.

or opening csrss.exe(420):
ess is denied.

or opening wininit.exe(512):
ess is denied.

or opening csrss.exe(520):
ess is denied.

-----
logon.exe pid: 608
mand line: winlogon.exe
```


e	Size	Path
000000ba140000	0xe3000	C:\Windows\system32\winlogon.exe
0000000a1e70000	0x1f8000	C:\Windows\SYSTEM32\ntdll.dll
0000000a16f0000	0xbd000	C:\Windows\System32\KERNEL32.DLL
00000009f910000	0x2f6000	C:\Windows\System32\KERNELBASE.dll
0000000a0c00000	0x9e000	C:\Windows\System32\msvcrt.dll
0000000a17f0000	0x9c000	C:\Windows\System32\sechost.dll
0000000a0360000	0x126000	C:\Windows\System32\RPCRT4.dll
0000000a1ad0000	0x354000	C:\Windows\System32\combase.dll
00000009fda0000	0x100000	C:\Windows\System32\ucrtbase.dll
0000000a02b0000	0xaf000	C:\Windows\System32\advapi32.dll
00000009f3b0000	0x4b000	C:\Windows\SYSTEM32\powrprof.dll
00000009f390000	0x12000	C:\Windows\system32\UMPDC.dll
00000009f480000	0x25000	C:\Windows\system32\profapi.dll
0000000a0a00000	0x19e000	C:\Windows\System32\user32.dll
00000009f6b0000	0x22000	C:\Windows\System32\win32u.dll
0000000a13f0000	0x2c000	C:\Windows\System32\GDI32.dll
00000009f7f0000	0x11a000	C:\Windows\System32\gdi32full.dll
00000009f750000	0x9d000	C:\Windows\System32\msvc_p_win.dll
0000000a1a40000	0x30000	C:\Windows\System32\IMM32.DLL
00000009f280000	0x5b000	C:\Windows\SYSTEM32\insta.dll
00000009f400000	0x32000	C:\Windows\system32\SspiCli.dll
00000009f440000	0x2e000	C:\Windows\system32\USERENV.dll
00000009e210000	0x30000	C:\Windows\SYSTEM32\profext.dll
00000009e670000	0x33000	C:\Windows\SYSTEM32\ntmarta.dll
00000009fea0000	0x27000	C:\Windows\System32\Bcrypt.dll
00000009fc60000	0x82000	C:\Windows\System32\bcryptprimitives.dll

00000009e910000	0x3b000	C:\Windows\SYSTEM32\IPHLPAPI.DLL
00000009fed0000	0x8000	C:\Windows\System32\NSI.dll
00000009e130000	0x36000	C:\Windows\system32\fwbase.dll
00000009cf00000	0x1e000	C:\Windows\SYSTEM32\uxinit.dll
0000000a0150000	0xad000	C:\Windows\System32\shcore.dll
00000009d300000	0x2f000	C:\Windows\system32\dwmapi.dll
00000009cf40000	0x9e000	C:\Windows\system32\UxTheme.dll
00000009f550000	0x15d000	C:\Windows\System32\CRYPT32.dll
00000009f270000	0xa000	C:\Windows\system32\DPAPI.dll
00000009ee30000	0xc000	C:\Windows\system32\CRYPTBASE.dll
00000009cdf0000	0x14000	C:\Windows\SYSTEM32\dwminit.dll
00000009cd60000	0x90000	C:\Windows\system32\apphelp.dll
00000009c9f0000	0x141000	C:\Windows\system32\dsreg.dll
0000000a1890000	0xcd000	C:\Windows\System32\OLEAUT32.dll
00000009e500000	0x8a000	C:\Windows\system32\msvc_p110_win.dll
00000009ee10000	0x18000	C:\Windows\SYSTEM32\cryptsp.dll
00000009a8c0000	0x16000	C:\Windows\SYSTEM32\usermgrcli.dll
00000009d410000	0x12000	C:\Windows\SYSTEM32\kernel.appcore.dll
000000090070000	0x1d000	C:\Windows\system32\MPR.dll
00000009e6b0000	0x19000	C:\Windows\system32\wksccli.dll
00000009ea20000	0xc000	C:\Windows\system32\netutils.dll

or opening services.exe(644):
ess is denied.

The **handle** command is an exceedingly extensive list of programs and processes that use specific files or other resources. This is another command that displays an extremely long output; therefore, I only included the beginning of it in the screenshot.

```

C:\Users\vboxuser\Documents\SysinternalsSuite>handle

NtHandle v5.0 - Handle viewer
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Sysinternals - www.sysinternals.com

-----
System pid: 4 \NT AUTHORITY\SYSTEM
-----
smss.exe pid: 324 \<unable to open process>
-----
csrss.exe pid: 420 \<unable to open process>
-----
wininit.exe pid: 512 \<unable to open process>
-----
csrss.exe pid: 520 \<unable to open process>
-----
winlogon.exe pid: 608 NT AUTHORITY\SYSTEM
  40: File (RW-) C:\Windows\System32
 1A8: Section \Sessions\1\Windows\Theme1038637390
 28C: Section \Windows\Theme3528996545
 290: File (R-D) C:\Windows\System32\en-US\winlogon.exe.mui
 2A8: Section \Sessions\1\Windows\ThemeSection
 3F0: File (R-D) C:\Windows\System32\en-US\user32.dll.mui
-----
services.exe pid: 644 \<unable to open process>
-----
lsass.exe pid: 664 NT AUTHORITY\SYSTEM
  40: File (RW-) C:\Windows\System32
 118: Section \LsaPerformance
 184: File (R-D) C:\Windows\System32\en-US\lsasrv.dll.mui
 28C: Section \BaseNamedObjects\Debug.Trace.Memory.298
 3EC: File (RW-) C:\Windows\debug\PASSWD.LOG
 B20: File (RWD) C:\Users\vboxuser\AppData\Roaming\Microsoft\Credentials
 D48: File (R-D) C:\Windows\System32\en-US\crypt32.dll.mui
 EA8: File (RWD) C:\Users\vboxuser\AppData\Local\Microsoft\Credentials
108C: File (R-D) C:\Windows\System32\en-US\vaultsvc.dll.mui
1100: Section \BaseNamedObjects\__ComCatalogCache__

```

Another way to collect process

information is to use the **pslist** command with its different options of **-d** (listing the thread details for Windows), **-m** (listing the process memory details for Windows), and **-x** (listing the process and thread details for Windows).

```

C:\Users\vboxuser\Documents\SysinternalsSuite>pslist -d

PsList v1.41 - Process information lister
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Sysinternals - www.sysinternals.com

Thread detail for WINDOWS:

Idle 0:
  Tid Pri   Cswtch      State      User Time   Kernel Time   Elapsed Time
   0   0    265439    Running   0:00:00.000  0:17:50.265  0:00:00.000
   0   0    285399    Running   0:00:00.000  0:14:08.000  0:00:00.000

System 4:
  Tid Pri   Cswtch      State      User Time   Kernel Time   Elapsed Time
  12  15       1  Wait:Executive  0:00:00.000  0:00:00.000  3726791:06:41.584
  16  13      10  Wait:Executive  0:00:00.000  0:00:00.000  3726791:06:41.584
  20  13      12  Wait:Executive  0:00:00.000  0:00:00.000  3726791:06:41.584
  24  16     6020  Wait:Executive  0:00:00.000  0:00:02.984  3726791:06:41.584
  28  12      268  Wait:Executive  0:00:00.000  0:00:00.328  2:01:38.040
  36   9    16693  Wait:Queue     0:00:00.000  0:00:23.437  2:01:38.037
  40  31       1  Wait:Suspended  0:00:00.000  0:00:00.000  2:01:38.033
  44  31       1  Wait:Suspended  0:00:00.000  0:00:00.000  2:01:38.032
  48  17     4727  Running        0:00:00.000  0:07:24.093  2:01:37.960
  52  23    40067  Wait:Executive  0:00:00.000  0:00:22.000  2:01:37.960
  56   8       31  Wait:Executive  0:00:00.000  0:00:00.000  2:01:37.960
  68   0    11576  Ready         0:00:00.000  0:00:44.687  2:01:37.959

```

```
C:\Users\vboxuser\Documents\SysinternalsSuite>pslist -m

PsList v1.41 - Process information lister
Copyright (C) 2000-2023 Mark Russinovich
Sysinternals - www.sysinternals.com

Process memory detail for WINDOWS:
```

Name	Pid	VM	WS	Priv	Priv	Pk	Faults	NonP	Page
Idle	0	8	8	60	60		9	0	0
System	4	3908	152	192	212		2229	0	0
Registry	92	133104	57164	2600	36964		78613	12	264
smss	324	4194303	960	1056	1124		984	3	12
csrss	420	4194303	4708	1736	1944		2842	14	167
wininit	512	4194303	6136	1372	1916		2248	10	75
csrss	520	4194303	5964	1976	2388		5704	19	193
winlogon	608	4194303	9352	2572	3652		6999	12	143
services	644	4194303	7220	3600	4288		4504	10	139
lsass	664	4194303	15872	7040	7632		10862	24	150
svchost	764	4194303	30028	10720	13456		17782	25	566
fontdrvhost	784	4194303	3940	1892	1892		1474	5	46
fontdrvhost	792	4194303	2360	1280	1308		999	5	36
svchost	876	4194303	14116	6964	8028		29778	18	151
dwm	988	4194303	62820	32376	38596		129145	44	483
svchost	368	4194303	67204	44164	75864	12175324		65	549
svchost	424	4194303	95812	82584	102500	1298291		26	195
svchost	720	4194303	22748	17072	20068	18985		19	153
svchost	704	4194303	21300	18688	35760	61429		17	139

```
C:\Users\vboxuser\Documents\SysinternalsSuite>pslist -x

PsList v1.41 - Process information lister
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Sysinternals - www.sysinternals.com

Process and thread information for WINDOWS:
```

Name	Pid	VM	WS	Priv	Priv	Pk	Faults	NonP	Page
Idle	0	8	8	60	60		9	0	0
Tid Pri Cswtch			State	User Time	Kernel Time		Elapsed Time		
0 0 279952			Running	0:00:00.000	0:19:59.640		0:00:00.000		
0 0 298071			Running	0:00:00.000	0:15:44.468		0:00:00.000		

Name	Pid	VM	WS	Priv	Priv	Pk	Faults	NonP	Page
System	4	3908	152	192	212		2229	0	0
Tid Pri Cswtch			State	User Time	Kernel Time		Elapsed Time		
12 15 1			Wait:Executive	0:00:00.000	0:00:00.000		3726791:08:42.972		
16 13 10			Wait:Executive	0:00:00.000	0:00:00.000		3726791:08:42.972		
20 13 12			Wait:Executive	0:00:00.000	0:00:00.000		3726791:08:42.972		
24 16 6269			Wait:Executive	0:00:00.000	0:00:02.984		3726791:08:42.972		
28 12 273			Wait:Executive	0:00:00.000	0:00:00.328		2:03:39.427		
36 9 17190			Wait:Queue	0:00:00.000	0:00:23.515		2:03:39.424		
40 31 1			Wait:Suspended	0:00:00.000	0:00:00.000		2:03:39.420		
44 31 1			Wait:Suspended	0:00:00.000	0:00:00.000		2:03:39.420		
48 17 4885			Wait:Executive	0:00:00.000	0:07:31.781		2:03:39.348		

Once we collect the process information, we can then examine the process memory. We can perform this task using the **procexp** command. After inputting this command, the Process Explorer window opens immediately.

Process Explorer - Sysinternals: www.sysinternals.com [WINDOWS\vboxuser] (Administrator)

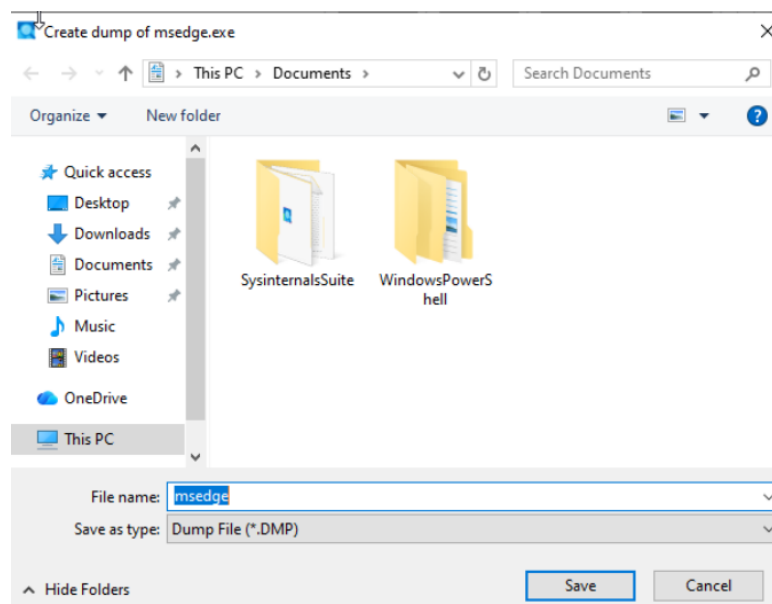
File Options View Process Find Users Help

<Filter by name>

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Registry		2,632 K	57,496 K	92		
System Idle Process	< 0.01	60 K	8 K	0		
System	13.96	192 K	152 K	4		
Interrupts	12.05	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,056 K	968 K	324		
Memory Compression	< 0.01	520 K	59,856 K	1404		
csrss.exe		1,732 K	4,756 K	420		
wininit.exe		1,372 K	6,144 K	512		
services.exe	7.61	3,672 K	7,344 K	644		
svchost.exe		10,664 K	30,032 K	764	Host Process for Windows S...	Microsoft Corporation
dlhst.exe		3,444 K	10,676 K	2292	COM Surrogate	Microsoft Corporation
TextInputHost.exe		8,928 K	35,976 K	2920		Microsoft Corporation
StartMenuExperienceHost.exe		18,828 K	54,352 K	4088		Microsoft Corporation
RuntimeBroker.exe		6,476 K	25,460 K	2232	Runtime Broker	Microsoft Corporation
dlhst.exe		3,584 K	12,296 K	5048	COM Surrogate	Microsoft Corporation
RuntimeBroker.exe		2,168 K	9,644 K	5000	Runtime Broker	Microsoft Corporation
MoUsocoreWorker.exe		26,052 K	28,720 K	2228	MoUsocore Worker Process	Microsoft Corporation
wuauclt.exe		13,276 K	24,904 K	7124	Windows Update	Microsoft Corporation
UserOOBEBroker.exe		2,004 K	9,356 K	4128	User OOBEBroker	Microsoft Corporation
ShellExperienceHost.exe	Susp...	13,936 K	43,848 K	5804	Windows Shell Experience H...	Microsoft Corporation
RuntimeBroker.exe		3,604 K	19,376 K	3680	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		10,184 K	33,776 K	4292	Runtime Broker	Microsoft Corporation
SearchApp.exe	Susp...	130,276 K	93,496 K	3884	Search application	Microsoft Corporation
Microsoft.Photos.exe	Susp...	26,332 K	36,220 K	2532		Microsoft Corporation
RuntimeBroker.exe		3,388 K	16,060 K	1836	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		2,640 K	14,228 K	8064	Runtime Broker	Microsoft Corporation
TIWorker.exe	19.03	454,056 K	389,212 K	4284	Windows Modules Installer ...	Microsoft Corporation
WmiPrvSE.exe		2,528 K	9,384 K	8048	WMI Provider Host	Microsoft Corporation
svchost.exe	1.27	7,060 K	14,216 K	876	Host Process for Windows S...	Microsoft Corporation

CPU Usage: 100.00% Commit Charge: 59.61% Processes: 92 Physical Usage: 76.83%

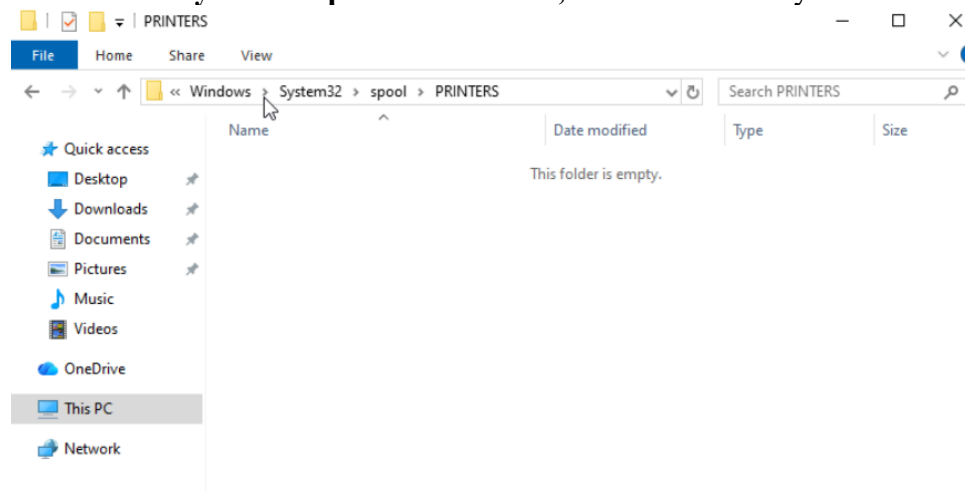
Within this window, we can create a full memory dump of a given process. To do this, I right clicked on the **msedge.exe** file and selected “Create Dump,” then “Create Full Dump.” Then the File Explorer window appears, and we can select the location of the dump file and save it.



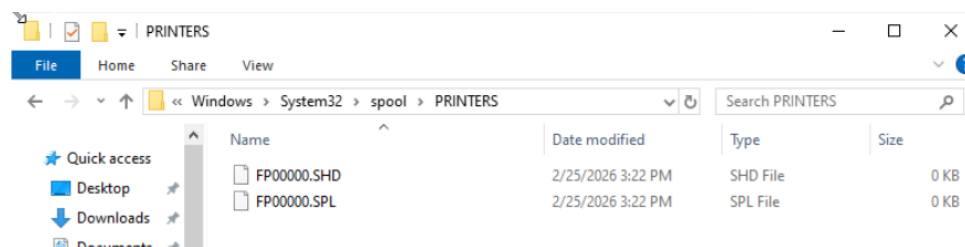
Another task we can complete within the Process Explorer window is to perform a malware analysis of a given service by right clicking on a file and selecting “Check VirusTotal.com.” This action adds a column to the Process Explorer window called “Virus Total”. The file we selected now shows the number of viruses within it.

Process	Private Bytes	Working Set	PID	Description	Company Name	VirusTotal
TrustedInstaller.exe	1,996 K	7,640 K	5964	Windows Modules Installer	Microsoft Corporation	
svchost.exe	1,624 K	8,380 K	12536			
lsass.exe	7,104 K	16,776 K	664	Local Security Authority Proc...	Microsoft Corporation	
fontdrvhost.exe	1,280 K	2,332 K	792	Usermode Font Driver Host	Microsoft Corporation	
csrss.exe	2,132 K	6,296 K	520			
winlogon.exe	2,572 K	9,260 K	608	Windows Logon Application	Microsoft Corporation	
fontdrvhost.exe	3,268 K	5,060 K	784	Usermode Font Driver Host	Microsoft Corporation	
dwm.exe	36,404 K	49,176 K	988	Desktop Window Manager	Microsoft Corporation	
explorer.exe	85,936 K	97,024 K	3528	Windows Explorer	Microsoft Corporation	
SecurityHealthSystray.exe	1,920 K	5,800 K	3520	Windows Security notificatio...	Microsoft Corporation	
cmd.exe	2,556 K	4,128 K	12228	Windows Command Processor	Microsoft Corporation	
conhost.exe	9,132 K	15,608 K	10988	Console Window Host	Microsoft Corporation	
procexp.exe	5,092 K	12,212 K	12980	Sysinternals Process Explorer	Sysinternals - www.sysinter...	
procexp64.exe	23,840 K	39,056 K	11356	Sysinternals Process Explorer	Sysinternals - www.sysinter...	
M365Copilot.exe	28,356 K	24,864 K	3608	Microsoft 365 Copilot App	Microsoft Corporation	
msedgewebview2.exe	36,144 K	25,424 K	4236	Microsoft Edge WebView2	Microsoft Corporation	
msedgewebview2.exe	2,264 K	3,776 K	892	Microsoft Edge WebView2	Microsoft Corporation	
msedgewebview2.exe	14,620 K	7,344 K	1640	Microsoft Edge WebView2	Microsoft Corporation	
msedgewebview2.exe	8,664 K	10,264 K	3880	Microsoft Edge WebView2	Microsoft Corporation	
msedgewebview2.exe	26,220 K	8,052 K	3108	Microsoft Edge WebView2	Microsoft Corporation	
msedgewebview2.exe	9,720 K	7,220 K	4420	Microsoft Edge WebView2	Microsoft Corporation	
OneDrive.exe	49,828 K	5,596 K	3840	Microsoft OneDrive	Microsoft Corporation	
OneDrive.Sync.Service.exe	39,192 K	3,136 K	6352	Microsoft OneDrive Sync Ser...	Microsoft Corporation	
msedge.exe	46,256 K	122,932 K	12244	Microsoft Edge	Microsoft Corporation	0/77
msedge.exe	2,316 K	8,580 K	9868	Microsoft Edge	Microsoft Corporation	0/77
msedge.exe	12,068 K	34,188 K	12988	Microsoft Edge	Microsoft Corporation	0/77
msedge.exe	11,988 K	25,436 K	12932	Microsoft Edge	Microsoft Corporation	0/77
msedge.exe	8,808 K	19,360 K	7108	Microsoft Edge	Microsoft Corporation	0/77

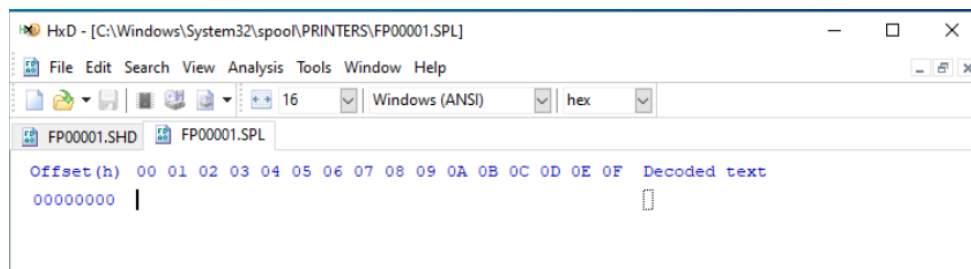
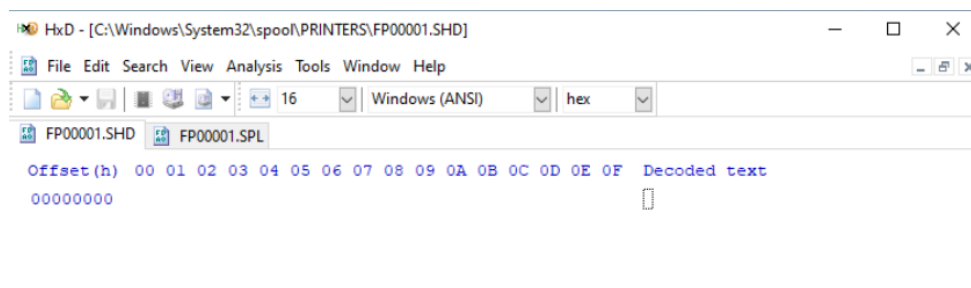
The last step of this lab focuses on examining the print spool. Windows stores printing tasks in a spool file (.spl) and a shadow file (.shd). When locating **C:\windows\system32\spool\PRINTERS**, there were not any files within this folder.



To generate files within this folder, I sent a print request for a picture that was saved to this virtual machine. Once the print request was made, two files appeared within the PRINTERS folder: one spool file and one shadow file.



I opened the HxD software and added the shadow and spool file to it. Unfortunately, the data for both files did not appear.



Part 3: References

1. Lab 6: Windows Forensics Part 1
2. Lecture notes: Windows Forensics – Part 1
3. [What is the Handle Command?](#)